

Continuum Mechanics

class # 09

9/15/2026

Functions of Second-Order Tensors

Definition (2.20): A function $\psi: \mathcal{V}^2 \rightarrow \mathbb{R}$ is said to be differentiable at $A \in \mathcal{V}^2$ iff there exists a second-order tensor $D\psi(A) \in \mathcal{V}^2$ such that

$$\psi(A+H) = \psi(A) + D\psi(A):H + o(|H|).$$

The tensor $D\psi(A)$ is called the derivative of ψ at A .

Proposition: If $\psi: \mathcal{V}^2 \rightarrow \mathbb{R}$ is differentiable at A then its derivative is unique and

$$D\psi(A):B = \left. \frac{d}{d\alpha} \psi(A+\alpha B) \right|_{\alpha=0}$$

for all $B \in \mathcal{V}^2$.

Proof: Any good book on Advanced Calculus. |||

Result (2.21): let $\mathcal{F} = \{\vec{e}_i\}$ be an arbitrary frame. Then

$$D\psi(A) = \frac{\partial \psi}{\partial A_{ij}}(A) \vec{e}_i \otimes \vec{e}_j$$

where

$$A = A_{ij} \vec{e}_i \otimes \vec{e}_j$$

in the frame $\mathcal{F}$.

Proof: let us write, in an abuse of notation

$$\psi(A) = \psi(A_{11}, A_{12}, \dots, A_{33}).$$

let $\alpha \in \mathbb{R}$ be any scalar and

$$B = B_{ij} \vec{e}_i \otimes \vec{e}_j$$

be any two tensor. Then

$$\psi(A + \alpha B) = \psi(A_{11} + \alpha B_{11}, \dots, A_{33} + \alpha B_{33}).$$

By the chain rule

$$\begin{aligned} D\psi(A): B &= \frac{d}{d\alpha} \psi(A + \alpha B) \Big|_{\alpha=0} \\ &= \frac{\partial \psi}{\partial A_{ij}}(A) B_{ij} \\ &= \left(\frac{\partial \psi}{\partial A_{ij}}(A) \vec{e}_i \otimes \vec{e}_j \right) : (B_{kl} \vec{e}_k \otimes \vec{e}_l) \\ &= \left(\frac{\partial \psi}{\partial A_{ij}}(A) \vec{e}_i \otimes \vec{e}_j \right) : B. \end{aligned}$$

Since B is arbitrary,

$$D\psi(A) = \frac{\partial \psi}{\partial A_{ij}}(A) \vec{e}_i \otimes \vec{e}_j. \quad //$$

Remark: The derivative of a scalar function $\psi: \mathcal{V}^2 \rightarrow \mathbb{R}$ is a second-order tensor-valued function. //

Proposition: Using Taylor's Theorem we can prove that

$$\begin{aligned}\Psi(A+H) &= \Psi(A) + \frac{\partial \Psi}{\partial A_{ij}}(A) H_{ij} + \mathcal{O}(|H|^2) \\ &= \Psi(A) + D\Psi(A):H + \mathcal{O}(|H|^2),\end{aligned}$$

provided that Ψ is smooth enough.

Proof: A good book on Advanced Calculus. //

Remark: The last result shows that for smooth functions $\Psi(A) + D\Psi(A):H$ is a better approximation for $\Psi(A+H)$ than what is indicated by the definition of $D\Psi(A)$. //

Example: Consider the scalar function $\Psi: \mathcal{V}^2 \rightarrow \mathbb{R}$ defined by

$$\Psi(A) = \frac{1}{2} A:A = \frac{1}{2} A_{kl} A_{kl}$$

where in the frame $\mathcal{F} = \{\vec{e}_i\}$

$$A = A_{ij} \vec{e}_i \otimes \vec{e}_j.$$

Then

$$\begin{aligned}\frac{\partial \Psi}{\partial A_{ij}}(A) &= \frac{1}{2} \frac{\partial A_{kl}}{\partial A_{ij}} A_{kl} + \frac{1}{2} A_{kl} \frac{\partial A_{kl}}{\partial A_{ij}} \\ &= A_{kl} \frac{\partial A_{kl}}{\partial A_{ij}} \\ &= A_{kl} \delta_{ik} \delta_{jl} \\ &= A_{ij}.\end{aligned}$$

Therefore, by Result (2.21),

$$D\psi(A) = A_{ij} \vec{e}_i \otimes \vec{e}_j = A.$$

Result (2.22): Let $\psi(S) = \det(S)$. If A is invertible, i.e., $\det(A) \neq 0$, then ψ is differentiable at A and

$$D\psi(A) = \det(A) A^{-T}.$$

Proof: Let $B \in V^2$ be arbitrary. Then

$$\psi(A + \alpha B) = \det(A + \alpha B)$$

$$= \det(\alpha A (A^{-1}B - \lambda I)) \quad (\lambda = -\frac{1}{\alpha})$$

$$= \det(\alpha A) \det(A^{-1}B - \lambda I)$$

$$= \alpha^3 \det(A) \det(A^{-1}B - \lambda I).$$

Recall that the characteristic polynomial satisfies

$$p_S(\lambda) = \det(S - \lambda I) = -\lambda^3 + I_1(S)\lambda^2 - I_2(S)\lambda + I_3(S)$$

$$I_1(S) = \text{tr} S$$

$$I_2(S) = \frac{1}{2}[(\text{tr} S)^2 - \text{tr}(S^2)]$$

$$I_3(S) = \det(S).$$

Hence

$$\begin{aligned}\psi(A + \alpha B) &= \det(A) + \alpha \det(A) I_1(A^{-1}B) \\ &\quad + \alpha^2 \det(A) I_2(A^{-1}B) \\ &\quad + \alpha^3 \det(A) I_3(A^{-1}B)\end{aligned}$$

$$\left. \frac{d}{d\alpha} \psi(A + \alpha B) \right|_{\alpha=1} = \det(A) I_1(A^{-1}B).$$

Finally

$$I_1(A^{-1}B) = \text{tr}(A^{-1}B) = A^{-T} : B.$$

Hence

$$D\psi(A) = \det(A) A^{-T}. \quad //$$

Exercise: Show that, for any $S, T \in \mathcal{V}^2$,

$$\text{tr}(ST) = S^T : T.$$

Proof: Let $\mathcal{F} = \{\vec{e}_i\}$ be a frame

$$\text{tr}(ST) = [ST]_{\ell\ell}$$

But

$$[ST]_{ij} = [S]_{ik} [T]_{kj}$$

So

$$\begin{aligned}[ST]_{\ell\ell} &= [S]_{\ell k} [T]_{k\ell} \\ &= [S^T]_{k\ell} [T]_{k\ell} \\ &= S^T : T \quad //\end{aligned}$$

Result (2.23): let S be a time-dependent second-order tensor, i.e., $S: \mathbb{R} \rightarrow \mathcal{V}^2$. Then

$$\begin{aligned} \frac{d}{dt}(\det(S)) &= \det(S) \operatorname{tr}(S^{-1} \dot{S}) \\ &= \det(S) S^{-T} : \dot{S}, \end{aligned}$$

where

$$\dot{S} := \frac{dS}{dt} = \frac{dS_{ij}}{dt} \vec{e}_i \otimes \vec{e}_j.$$

Proof: Set $\psi(S) = \det(S)$. Write, as usual,

$$\psi(S) = \psi(S_{11}, S_{12}, \dots, S_{33})$$

Thus,

$$\frac{d}{dt} \psi(S) = \frac{d}{dt} \psi(S_{11}, S_{12}, \dots, S_{33})$$

By the chain rule,

$$\begin{aligned} \frac{d}{dt} \psi(S) &= \frac{\partial \psi}{\partial S_{ij}}(S) \frac{dS_{ij}}{dt} \\ &= D\psi(S) : \dot{S}. \end{aligned}$$

But, in the previous result, we learned that

$$D\psi(S) = \det(S) S^T$$

So,

$$\frac{d}{dt}(\det(S)) = \det(S) S^{-T} : \dot{S}. \quad \text{///}$$

Tensor-Valued Functions

Definition (2.24): A function $T: V^2 \rightarrow V^2$ is said to be differentiable at $A \in V^2$ iff there exists a fourth-order tensor $DT(A) \in V^4$ such that

$$T(A+H) = T(A) + DT(A)H + o(|H|).$$

The fourth-order tensor $DT(A)$ is called the derivative of T at A .

Proposition: If $T: V^2 \rightarrow V^2$ is differentiable at $A \in V^2$ the derivative is unique and

$$DT(A)B = \left. \frac{d}{d\alpha} T(A + \alpha B) \right|_{\alpha=0}$$

for all $B \in V^2$.

Result (2.25): let $\mathcal{F} = \{\vec{e}_i\}$ be any frame and Suppose

$$T(A) = T_{ij}(A) \vec{e}_i \otimes \vec{e}_j.$$

Then, if T is differentiable at $A \in V^2$,

$$DT(A) = \frac{\partial T_{ij}}{\partial A_{kl}}(A) e_i \otimes e_j \otimes \vec{e}_k \otimes \vec{e}_l,$$

where

$$A = A_{ij} \vec{e}_i \otimes \vec{e}_j.$$

Proof: We use our standard abuse of notation:

$$T_{ij}(A) = T_{ij}(A_{11}, A_{12}, \dots, A_{33})$$

For any $\alpha \in \mathbb{R}$ and $B = B_{ij} \vec{e}_i \otimes \vec{e}_j \in V^2$, we have

$$T_{ij}(A + \alpha B) = T_{ij}(A_{11} + \alpha B_{11}, A_{12} + \alpha B_{12}, \dots, A_{33} + \alpha B_{33}).$$

By the chain rule

$$\frac{d}{d\alpha} T_{ij}(A + \alpha B)|_{\alpha=0} = \frac{\partial T_{ij}}{\partial A_{kl}}(A) B_{kl}.$$

By the last proposition

$$\begin{aligned} DT(A)B &= \frac{d}{d\alpha} T(A + \alpha B)|_{\alpha=0} \\ &= \frac{d}{d\alpha} \left(T_{ij}(A + \alpha B) \vec{e}_i \otimes \vec{e}_j \right) \Big|_{\alpha=0} \\ &= \left(\frac{d}{d\alpha} T_{ij}(A + \alpha B) \Big|_{\alpha=0} \right) \vec{e}_i \otimes \vec{e}_j \\ &= \frac{\partial T_{ij}}{\partial A_{kl}}(A) B_{kl} \vec{e}_i \otimes \vec{e}_j. \end{aligned}$$

Now recall that in Chapter 1 we showed that if $C \in V^4$ and $U, T \in V^2$ satisfy

$$U = CT,$$

then

$$\begin{aligned} U &= (C_{ijkl} \vec{e}_i \otimes \vec{e}_j \otimes \vec{e}_k \otimes \vec{e}_l) T \\ &= C_{ijkl} T_{kl} \vec{e}_i \otimes \vec{e}_j \end{aligned}$$

where

$$T = T_{kl} \vec{e}_k \otimes \vec{e}_l$$

and

$$C = C_{ijkl} \vec{e}_i \otimes \vec{e}_j \otimes \vec{e}_k \otimes \vec{e}_l.$$

The scalars C_{ijkl} are the components of $C \in V^4$ with respect to the frame $\mathcal{F} = \{\vec{e}_i\}$.

Therefore,

$$\begin{aligned} DT(A) B &= \frac{\partial T_{ij}}{\partial A_{kl}}(A) B_{kl} \vec{e}_i \otimes \vec{e}_j \\ &= \left(\frac{\partial T_{ij}}{\partial A_{kl}} \vec{e}_i \otimes \vec{e}_j \otimes \vec{e}_k \otimes \vec{e}_l \right) B. \end{aligned}$$

Since $B \in V^2$ is arbitrary,

$$DT(A) = \frac{\partial T_{ij}}{\partial A_{kl}} \vec{e}_i \otimes \vec{e}_j \otimes \vec{e}_k \otimes \vec{e}_l. \quad ///$$

Remark: The derivative of a second-order tensor-valued function $T: V^2 \rightarrow V^2$ is a fourth-order tensor-valued function $DT: V^2 \rightarrow V^4$.

Proposition: If $T: V^2 \rightarrow V^2$ is sufficiently smooth then Taylor's Theorem implies that

$$T(A+H) = T(A) + DT(A)H + \mathcal{O}(\|H\|^2).$$

Example: Suppose that $T: V^2 \rightarrow V^2$ is defined as

$$T(A) = \text{tr}(A) A.$$

Then,

$$T_{ij}(A) = \text{tr}(A) A_{ij} = A_{mm} A_{ij}$$

The components of the derivative four-tensor are

$$[DT(A)]_{ijkl} = \frac{\partial T_{ij}}{\partial A_{kl}}(A)$$

Thus

$$\begin{aligned} [DT(A)]_{ijkl} &= \frac{\partial}{\partial A_{kl}} (A_{mm} A_{ij}) \\ &= \frac{\partial A_{mm}}{\partial A_{kl}} A_{ij} + A_{mm} \frac{\partial A_{ij}}{\partial A_{kl}} \\ &= \delta_{km} \delta_{lm} A_{ij} + A_{mm} \delta_{ki} \delta_{lj} \\ &= \delta_{kl} A_{ij} + A_{mm} \delta_{ik} \delta_{jl}. \end{aligned}$$

For any $B \in \mathcal{V}^2$

$$\begin{aligned} DT(A) B &= (\delta_{kl} A_{ij} + A_{mm} \delta_{ik} \delta_{jl}) B_{kl} \vec{e}_i \otimes \vec{e}_j \\ &= (B_{kk} A_{ij} + A_{mm} B_{ij}) \vec{e}_i \otimes \vec{e}_j \\ &= \text{tr}(B) A + \text{tr}(A) B \quad // \end{aligned}$$

Example (Exercise 2.15) : Suppose that

$$\Psi(A) = \text{tr}(A) A : B$$

where $B \in \mathcal{V}^2$ is a constant second-order tensor.
Show that

$$D\Psi(A) = (A : B)I + \text{tr}(A)B.$$

Recall

$$D\Psi(A) = \frac{\partial \Psi}{\partial A_{ij}}(A) \vec{e}_i \otimes \vec{e}_j$$

where $\mathcal{F} = \{\vec{e}_i\}$ is a frame and

$$A = A_{ij} \vec{e}_i \otimes \vec{e}_j.$$

Now,

$$\psi(A) = A_{mm} A_{kl} B_{kl}$$

So

$$\begin{aligned} \frac{\partial \psi}{\partial A_{ij}}(A) &= \frac{\partial}{\partial A_{ij}} (A_{mm} A_{kl} B_{kl}) \\ &= B_{kl} \left(\frac{\partial A_{mm}}{\partial A_{ij}} A_{kl} + A_{mm} \frac{\partial A_{kl}}{\partial A_{ij}} \right) \\ &= B_{kl} (\delta_{im} \delta_{jm} A_{kl} + A_{mm} \delta_{ik} \delta_{jl}) \\ &= B_{kl} (\delta_{ij} A_{kl} + A_{mm} \delta_{ik} \delta_{jl}) \\ &= (A_{kl} B_{kl}) \delta_{ij} + A_{mm} B_{ij} \end{aligned}$$

We know that

$$\begin{aligned} D\psi(A) &= \frac{\partial \psi}{\partial A_{ij}}(A) \vec{e}_i \otimes \vec{e}_j \\ &= [(A_{kl} B_{kl}) \delta_{ij} + A_{mm} B_{ij}] \vec{e}_i \otimes \vec{e}_j \\ &= (A:B) \mathbf{I} + \text{tr}(A) B. \quad /// \end{aligned}$$